

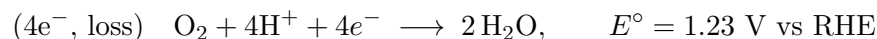
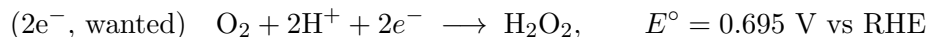
Equation Reference — Research Presentation (May 27)

Each card maps to a slide; screenshot it directly onto that slide. Forms reflect the current production stack: $\text{K}^+/\text{SO}_4^{2-}$ analytic Bikerman counterions, parallel $2\text{e}^-/4\text{e}^-$ ORR, Stern Robin BC ($C_S = 0.20 \text{ F/m}^2$), log- c unknowns with proton electrochemical potential μ_H , log-rate Butler–Volmer.

Symbols: F Faraday constant, R gas constant, T temperature, $V_T = RT/F$ thermal voltage; D_i, z_i, c_i diffusivity, charge, concentration of species i ; φ electrostatic potential; ε permittivity. Dynamic species $\{\text{O}_2, \text{H}_2\text{O}_2, \text{H}^+\}$ ($z = 0, 0, +1$); counterions $\{\text{K}^+, \text{SO}_4^{2-}\}$ carried analytically.

Slide 5 — ORR selectivity: two competing pathways

We want the 2e^- product (H_2O_2), not the 4e^- loss to water.



Selectivity = keeping O_2 on the 2e^- branch. The two reactions run *in parallel* at the electrode (Ruggiero 2022 topology).

Slide 6 — RRDE measurement & transport

Rotating ring–disk electrode: the disk makes H_2O_2 , the ring downstream detects it.

$$\text{collection efficiency} \quad N = 0.224 \quad (\text{Ruggiero 2022})$$

Rotation sets the diffusion-layer thickness via the Levich relation (the physical meaning of the domain length L_{eff}):

$$L_{\text{eff}} \approx 1.61 D_i^{1/3} \nu^{1/6} \omega^{-1/2}, \quad \omega = \frac{2\pi \Omega}{60},$$

with ν the kinematic viscosity and Ω the rotation rate (rpm). For O_2 in water this gives $L_{\text{eff}} \approx 86, 43, 35 \text{ }\mu\text{m}$ at $\Omega = 400, 1600, 2500 \text{ rpm}$; the model uses $L_{\text{eff}} \approx 100 \text{ }\mu\text{m}$.

Slide 12 — The PNP–BV model (governing equations + every BC)

Nernst–Planck in the diffuse layer, one equation per dynamic species i :

$$\frac{\partial c_i}{\partial t} = \nabla \cdot \left(D_i \left[\nabla c_i + \frac{z_i F}{RT} c_i \nabla \varphi \right] \right) = -\nabla \cdot \mathbf{J}_i, \quad \mathbf{J}_i = -D_i \left(\nabla c_i + \frac{z_i F}{RT} c_i \nabla \varphi \right).$$

Poisson for the potential (counterions enter as an analytic source — next card):

$$-\nabla \cdot (\varepsilon \nabla \varphi) = \sum_i z_i F c_i.$$

Boundary conditions.

- *Electrode / OHP — Butler–Volmer flux* (species i , reactions r):

$$-D_i \left(\nabla c_i + \frac{z_i F}{RT} c_i \nabla \varphi \right) \cdot \mathbf{n} = \sum_r \nu_{i,r} R_r,$$

$$R_r = k_{0,r} \left[c_O e^{-\alpha_r F \eta_r / RT} - c_R e^{(1-\alpha_r) F \eta_r / RT} \right], \quad \eta_r = \varphi_m - \varphi_s - E_{,r}^\circ.$$

Here c_O, c_R are the oxidized / reduced surface concentrations for reaction r and $\nu_{i,r}$ the stoichiometric coefficient ($E_{,r}^\circ \in \{0.695, 1.23\}$ V). The log-rate construction of this same rate law is on the Slide-14 card.

- *Stern compact layer — Robin BC on φ* :

$$\varepsilon \partial_n \varphi = C_S (V_{\text{app}} - \varphi_{\text{OHP}}), \quad C_S = 0.20 \text{ F/m}^2.$$

- *Bulk — Dirichlet*: $c_i = c_i^{\text{bulk}}, \quad \varphi = 0.$

Slide 12 (cont.) — Analytic Bikerman counterions & steric crowding

Non-reactive ions $b \in \{\text{K}^+, \text{SO}_4^{2-}\}$ are *not* discretized; in local equilibrium with φ they enter Poisson analytically.

Ideal (point-ion) Boltzmann limit:

$$c_b(x) = c_b^{\text{bulk}} \exp(-z_b F \varphi / RT).$$

Steric-aware Bikerman closure (caps packing at $c_b \leq 1/a_b$):

$$c_b(x) = c_b^{\text{bulk}} \frac{(1 - \sum_{j \in \text{dyn}} a_j c_j(x)) e^{-z_b F \varphi / RT}}{\theta_b + a_b c_b^{\text{bulk}} e^{-z_b F \varphi / RT}}, \quad \theta_b = 1 - \sum_j a_j c_j^{\text{bulk}} - a_b c_b^{\text{bulk}}.$$

Finite ion size (a_i = ion volume) adds a steric chemical potential to every species' driving force:

$$\mu_i^{\text{steric}} = -\ln \theta, \quad \theta = 1 - \sum_i a_i c_i, \quad c_{\text{max}} \approx \frac{1}{a} \Rightarrow \mathbf{J}_i = -D_i c_i \left(\nabla u_i + \frac{z_i F}{RT} \nabla \varphi + \nabla \mu_i^{\text{steric}} \right).$$

Slide 13 — Why it is numerically hard

In equilibrium each ion piles up (or depletes) Boltzmann-exponentially at the interface:

$$c_i \approx c_i^{\text{bulk}} \exp(-z_i F \varphi / RT) \implies \text{concentrations span } \sim 10 \text{ orders of magnitude.}$$

The drop occurs across a Debye length that is tiny next to L_{eff} :

$$\lambda_D = \sqrt{\frac{\varepsilon RT}{2F^2 I}} \ll L_{\text{eff}}, \quad I = \frac{1}{2} \sum_i z_i^2 c_i \text{ (ionic strength).}$$

The exponential BV nonlinearity at the boundary feeds this buildup, and electromigration vs. diffusion run on very different timescales — together a stiff, ill-conditioned system.

Slide 14 — How we solve it (I): log- c and proton μ_H

Log-concentration unknowns. Solve for $u_i = \ln c_i$, not c_i ; reconstruct $c_i = e^{u_i} > 0$ (positive by construction, tames the buildup):

$$u_i = \ln c_i, \quad \mathbf{J}_i = -D_i e^{u_i} \left(\nabla u_i + \frac{z_i F}{RT} \nabla \varphi \right).$$

Proton electrochemical potential. For H^+ use

$$\boxed{\mu_H = u_H + \frac{z_H F}{RT} \varphi} \quad (z_H = +1), \quad c_H = \exp\left(\mu_H - \frac{z_H F}{RT} \varphi\right), \quad \mathbf{J}_H = -D_H c_H \nabla \mu_H.$$

In local Boltzmann equilibrium $\nabla \mu_H \approx 0$, so the primary unknown μ_H stays *smooth across the double layer* even though u_H and φ each swing by tens of log-units.

Slide 14 — Butler–Volmer: general form vs. log-rate construction

General Butler–Volmer boundary equation (per reaction r ; the Slide-12 form), with c_O, c_R the oxidized / reduced surface concentrations:

$$R_r = k_{0,r} \left[c_O e^{-\alpha_r F \eta_r / RT} - c_R e^{(1-\alpha_r) F \eta_r / RT} \right], \quad \eta_r = \varphi_m - \varphi_s - E_{r,s}^\circ,$$

imposed through the flux BC $-D_i (\nabla c_i + \frac{z_i F}{RT} c_i \nabla \varphi) \cdot \mathbf{n} = \sum_r \nu_{i,r} R_r$.

Log-rate construction. With log-concentration unknowns $c_O = e^{u_O}$, $c_R = e^{u_R}$, fold each concentration *inside* its exponent, build each branch in log-space, and exponentiate once:

$$\begin{aligned} \log r_r^{(\text{cat})} &= \ln k_{0,r} + u_O - \alpha_r \frac{F \eta_r}{RT}, & r_r^{(\text{cat})} &= \exp(\log r_r^{(\text{cat})}), \\ \log r_r^{(\text{anod})} &= \ln k_{0,r} + u_R + (1 - \alpha_r) \frac{F \eta_r}{RT}, & r_r^{(\text{anod})} &= \exp(\log r_r^{(\text{anod})}), \\ R_r &= r_r^{(\text{cat})} - r_r^{(\text{anod})}. \end{aligned}$$

The contrast. Both give an *identical* R_r in exact arithmetic. The general form evaluates $c_O = \exp(\text{clamp } u_O)$ then multiplies by the BV exponential, so when u_O is very negative the clamp pins a finite floor and the rate cannot vanish (a phantom current). The log-rate form keeps u_O inside the exponent, so $\log r_r \rightarrow -\infty$ smoothly and the branch dies off correctly — same equation, numerically robust.

Slide 15 — Continuation: deform easy → hard along three knobs

Never solve the hard problem cold. Ramp three parameters from an easy regime to the physical target, warm-starting Newton at every step:

knob	start	target	effect
reaction rate k_0	10^{-12}	1 (physical)	turn the reaction on gradually
Stern capacitance C_S	0.10	0.20 F/m ²	stiffen the compact layer
voltage V_{RHE}	-0.5	+1.0 V	walk the operating window
solution at step $k + 1$: $\mathbf{U}_{k+1} = \text{Newton}(\mathbf{U}_k \text{ as initial guess})$.			

Slide 16 — When a step fails: recover, don't restart

If a continuation rung fails between a converged value a and a failed target b , insert the **geometric** ($\sqrt{}$) **midpoint** and retry from the last good solution:

$$k_{\text{mid}} = \sqrt{a b}.$$

Recurse this bisection to walk across the Frumkin cliff (up to depth 5; 8 warm-walk substeps), rebuilding nothing — only the parameter moves.

Slide 17 — Verification: method of manufactured solutions

Pick a smooth exact solution u^* , insert it into the operator $\mathcal{L}$ to get a source $f = \mathcal{L}u^*$, solve $\mathcal{L}u_h = f$, and measure the error as the mesh refines ($h \rightarrow 0$):

$$\|u^* - u_h\|_{L^2} = \mathcal{O}(h^{p+1}), \quad \|u^* - u_h\|_{H^1} = \mathcal{O}(h^p).$$

For CG1 elements ($p = 1$) the expected slopes are $L^2 \approx 2$, $H^1 \approx 1$, and the observed rates match for every field (u_{O_2} , $u_{\text{H}_2\text{O}_2}$, μ_H , φ) on the production logc.muh + Cs⁺/SO₄²⁻ + Stern stack.

Slide 18 — Agreement with prior work (closure reproduction)

To compare against the single-reaction closure model we (i) run one 2e⁻ reaction and (ii) align the open-circuit-potential convention by shifting the *whole* voltage axis and *both* equilibrium potentials by the same ΔV :

$$V_{\text{RHE}} \rightarrow V_{\text{RHE}} - \Delta V, \quad E_{,r}^\circ \rightarrow E_{,r}^\circ - \Delta V, \quad \Delta V = V_{\text{OCP}} = 0.785 \text{ V (pH 2)}.$$

The overpotential is invariant, so kinetics are unchanged:

$$\eta_r = (V_{\text{RHE}} - \Delta V) - (E_{,r}^\circ - \Delta V) = V_{\text{RHE}} - E_{,r}^\circ, \quad V_{\text{OCP}} = 0.47 + 0.197 + 0.059 \text{ pH}.$$

Profiles are reconstructed from the muh unknown: $c_H = \exp(\mu_H - \frac{z_H F}{RT} \varphi)$. **Result:** 25/25 converge; O₂-ratio and profiles within $\sim 5\%$.

Slide 20 — Tentative: cation hydrolysis (field-shifted $\text{p}K_a$)

A polarized interface turns the hydrated cation into a *local proton source* — reachable by the general solver, not the closure:



The acidity shifts with the interfacial field (Singh 2016, Eq. 3–4):

$$\text{p}K_a(\sigma) = \text{p}K_a^{\text{bulk}} + \Delta\text{p}K_a(\sigma), \quad \text{p}K_a^{\text{bulk}} = B - \frac{A z^2}{r_{M-O}},$$

$$\Delta\text{p}K_a(\sigma) = 2A z \sigma r_{H-El} \left(1 - \frac{r_{M-O}^2}{r_{H-El}^2} \right), \quad \sigma = C_S (\varphi_m - \varphi_s) \text{ (local Stern charge)}.$$

Constants $A = 620.32$ pm, $B = 17.154$, $r_O = 63$ pm; the new surface proton flux is gated by a Langmuir-capped cation coverage Γ .